

Weitong Liu

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PERSONAL INFORMATION

Name: Weitong Liu

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EDUCATION

Beihang University, Beijing, China

Ph.D. in Engineering Thermophysics, Research Institute of Aero-Engine

Sep. 2021 – Jun. 2026 (expected)

GPA: 3.70 / 4.00

Supervisors: Prof. Guoqiang Xu and Prof. Yanchen Fu

Beihang University, Beijing, China

B.Eng. in Aerospace Propulsion Engineering, School of Energy and Power Engineering

Sep. 2017 – Jun. 2021

GPA: 3.82 / 4.00

HONORS & AWARDS

- **Shen Yuan Medal** Nov. 2025
Description: The highest student honor at Beihang University, awarded to 10 Ph.D. candidates annually (Top 0.1%).
- **Doctoral Student Special Program of the Young Elite Scientists Sponsorship Program** Dec. 2024
Description: National-level talent program by Chinese Association for Science and Technology; selected as one of the 31 candidates in Chinese Society of Astronautics.
- **First Prize of Scientific Research Outstanding Achievement Award** Dec. 2025
Awarding Body: Ministry of Education of China. A prestigious national-level award recognizing the top research achievements in Chinese universities.
- **ESI Highly Cited Paper Award** Jul. 2025
Description: Recognized for the paper published in *Energy* (Top 1% citations in the field of Engineering).
- Chinese National Scholarship for Doctoral Postgraduates (Top 3%) Dec. 2024
- The Academic Excellence Foundation of Beihang University for Ph.D. Students (Top 1%) Jun. 2025
- Excellent Academic Innovation Achievement Award, Beihang University (Top 10%) Jul. 2024
- Outstanding Postgraduate, Beihang University (Top 10%) Apr. 2024
- Merit Student, Beihang University (Top 10%) Nov. 2024
- Merit Student, Beihang University (Top 10%) Nov. 2025
- The First Prize of Academic Scholarship, Beihang University (Top 10%) 2024, 2025
- Graduate Freshmen Scholarship, Beihang University (Top 8%) Sep. 2021

PUBLICATION METRICS & HIGHLIGHTS

- **Track Record:** 15 Peer-reviewed papers (11 SCI-indexed), including 11 as First/Primary Author. 1 U.S. Patent Application.
- **Impact:** 1 ESI Highly Cited Paper (First Author published in *Energy*; Top 1% citations in Engineering).
- **Metrics:** >120 Citations | h-index: 7 (Source: Google Scholar).
- **Key Venues:** *Energy* (IF=9.4), *Int. J. Heat Mass Transfer* (IF=5.8), *Int. Commun. Heat Mass Transfer* (IF=6.4).

RESEARCH EXPERIENCE

Thermal Management System for Hydrogen-Fueled Aero-Engines

Oct. 2024 – Present

- **Architectural Innovation:** Proposed a synergistic heat recovery–dissipation architecture for liquid hydrogen (LH₂) turbofans, incorporating a helium-based intermediate cycle to achieve deep utilization of the LH₂ heat sink.
- **Methodological Breakthrough:** Developed an energy-flow modeling framework via Kirchhoff's law, significantly reducing computational redundancy while maintaining high prediction accuracy.

- **Optimization:** Established a multi-parameter thermodynamic analysis method to identify optimal operating boundaries, maximizing the engine's overall thermal efficiency.

Intermediate Cycle Heat Exchange System (Ph.D. Thesis)

Feb. 2023 – Present

- **Holistic Optimization:** Established a "design-to-operation" optimization framework for intermediate cycle systems, successfully reconciling the trade-off between system weight reduction and heat transfer performance enhancement.
- **System Modeling:** Developed a transfer matrix-based model to decouple complex heat transfer characteristics, achieving precise identification of system behaviors under variable working conditions.
- **Experimental Validation:** Led the construction of a multilevel, multi-fluid experimental rig. Conducted comprehensive tests to reveal the energy transport mechanisms of the intermediate cycle heat exchange system.

Experimental Study of Airfoil-Fin Printed Circuit Heat Exchanger (PCHE)

Oct. 2022 – Jan. 2023

- **Theoretical Contribution:** Derived new empirical correlations for Nusselt number and friction coefficient based on experimental data, providing a critical reference for the design of compact heat exchangers in advanced energy systems.
- **High-Pressure Experiment:** Evaluated the thermal-hydraulic performance of Airfoil-Fin PCHE under extreme conditions using supercritical hydrocarbon fuel and high-pressure water.

Heat Transfer of Supercritical Hydrocarbon Fuel

Jul. 2021 – May. 2023

- **Theoretical Contribution:** Established a unified correlation framework for supercritical mixed convection, quantitatively decoupling the competitive contributions of buoyancy and inertial forces.
- **Mechanism Discovery:** Revealed the vibration-enhanced heat transfer mechanisms of supercritical pressure hydrocarbon fuel in laminar flow, revealing the coupling effects of flow dynamics and thermal properties.
- **Property Measurement:** Designed a high-precision flow-method experimental setup to measure the density of aviation kerosene under supercritical pressures (6–8 MPa) and wide temperature ranges (323–783 K).

RESEARCH GRANTS & PROJECT MANAGEMENT

Project Leadership & Management

- **Technical Lead:** Independently managed **4** funded research projects involving complex experimental rig construction and validation.
- **Scope:** Directed the full project lifecycle for the Intermediate Cycle Heat Exchange System and PCHEs, from conceptual design to final acceptance.
- **Management:** Oversaw a total research budget of approx. **¥5.5 Million**; coordinated with 6 external manufacturing vendors to ensure timely delivery of critical components, including custom heat exchangers, valves, a helium compressor, pumps, and storage units.
- **Key Achievement:** Built a multilevel, multi-fluid experimental rig from scratch, ensuring high-pressure and high-temperature tests; delivered 15 comprehensive technical reports to the funding agencies.

Funding Acquisition Contributions

- **Principal Investigator:** Independently secured the **Academic Excellence Foundation of BUAA** for PhD Students.
- **Proposal Co-Drafter:** Contributed to the successful application of the **Fundamental Research Funds for the Central Universities** (No. 501XTCX2023146001 and 501QYZX2023146001), and **Beijing Nova Program** (No. 20240484560).
- **Contribution:** Formulated the research methodology for the design, analysis, and optimization of the hydrogen intermediate cycle heat exchange system; conducted the feasibility analysis and budget planning.

Mentorship

- **Student Supervisor:** Supervised 4 master students and 3 undergraduates on heat exchanger design, thermal system modeling, heat exchanger experiments, and PCHE simulation.
- **Outcome:** Resulted in 3 collaborative papers published in *J. Enhanc. Heat Transf.*, *IGTC 2023*, and *ACTS 2024*.

ACADEMIC ACTIVITIES & SERVICE

International Conference Attendance

- **Oral presentation:** "Innovative Thermal Management for Liquid Hydrogen Aero Engines: A Comparison with Kerosene Systems", 11th European Conference for Aerospace Sciences, *EUCASS 2025*, Rome, Italy, Jul. 2025.
- **Oral presentation:** "Experimental Insights into Thermal-Hydraulic Performance of A Compact Printed Circuit Heat Exchanger with Airfoil Fins Using High-Pressure Water", 7th ASME International Conference of Micro/Nanoscale Heat and Mass Transfer, *ASME MNHMT 2024*, Nottingham, UK, Aug. 2024.
- **Participant:** International Gas Turbine Congress, *IGTC 2023*, Kyoto, Japan, Nov. 2023.

Professional Service

- **Teaching Assistant:** Engineering Thermodynamics, Beihang University, Beijing, China, Fall 2022.
- **Co-Reviewer Experience:** Assisted supervisor Prof. Fu in conducting peer reviews for journals including International Journal of Heat and Mass Transfer, Applied Thermal Engineering, and International Journal of Thermal Sciences, providing detailed technical comments and editorial recommendations.

COMPLETE PUBLICATION LIST

A) Peer-Reviewed Journal Articles

1. **Liu W.**, Xu G., Gu X., Yao J., Li M., Lei M., Chen Q.*, Fu Y.*, "Experimental Analysis and Thermodynamic Modeling for Multilevel Heat Exchange System with Multifluid in Aero Engines", *Energy*, 2025, 315: 134373. **(ESI Highly Cited Paper)**
2. **Liu W.**, Xu G., Gu X., Liu Y., Wang J., Zhang J., Fu Y.*, "Synergistic Heat Recovery–Dissipation Architecture for Hydrogen Turbofans: Integrated Heat Current Modeling with Multi-Parameter Thermodynamic Analysis", *Energy*, 2025, 338: 138735.
3. **Liu W.**, Xu G., Liu Y., Gu X., Wang J., Zhang J., Fu Y.*, "From Design to Operation: Integrated Optimization of Intermediate Cycle Heat Exchange Systems for Aero Engines", *International Journal of Heat and Mass Transfer*, 2026, 256: 127904.
4. **Liu W.**, Xu G., Fu Y.*, Wen J., Zhang N., "Numerical Investigation on Forced, Natural, and Mixed Convective Heat Transfer of N-Decane in Laminar Flow at Supercritical Pressures", *International Journal of Heat and Mass Transfer*, 2023, 209: 124129.
5. Fu Y., **Liu W.**, Qi H., Chen Q., Wen J., Xu G.*, "Heat Transfer Area Optimization of Intermediate Heat-Exchange Cycle System for Aero Engines", *International Journal of Heat and Mass Transfer*, 2024, 220: 124995. **(Second author; primary contributing student author)**
6. **Liu W.**, Xu G., Gang X., Qi H., Li M., Wen J., Fu Y.*, "Theoretical Modeling, Experimental Validation, and Thermodynamic Analysis on Intermediate Heat-Exchange Cycle System", *International Communications in Heat and Mass Transfer*, 2024, 156: 107635.
7. **Liu W.**, Xu G., Zhi H., Wang R., Li M., Fu Y.*, "Experimental Evaluation of Hydrothermal Performance in Airfoil-Fin PCHE with Supercritical Pressure Hydrocarbon Fuel", *International Communications in Heat and Mass Transfer*, 2024, 159: 108279.
8. Fu Y., **Liu W.**, Wang J., Zhang L., Wen J., Wu H., Xu G.*, "Experimental Investigation on Heat Transfer Enhancement of Supercritical Pressure Aviation Kerosene in Tubular Laminar Flow by Vibration", *Applied Thermal Engineering*, 2024, 257: 124206. **(Second author; primary contributing student author)**
9. Fu Y., **Liu W.**, Shi S., Wang R., Liu Y., Xu G.*, "Density Measurements of Aviation Kerosene RP-3 Over the Temperature Range From (323 to 783 K) Under Supercritical Pressures (6 to 8 MPa)", *Chinese Journal of Aeronautics*, 2025, 38(7): 103474. **(Second author; primary contributing student author)**
10. Qi H., Xu G., **Liu W.**, Zhang L., Fu Y.*, "Flow and Heat Transfer Characteristics in Small Diameter Tube Bundles with A Staggered Layout: An Experimental Study", *Journal of Enhanced Heat Transfer*, 2024, 31(5): 33-52.
11. Zhang Z., Wu Z.*, Luo X., **Liu W.**, "Numerical Study on Convective Heat Transfer of Liquid Metal Gallium in Turbine Guide Vane", *Aerospace*, 2023, 10(6):548.

B) Conference Proceedings

1. **Liu W.**, Zhi H., Qi H., Fu Y.*, "Experimental Insights into Thermal-Hydraulic Performance of A Compact Printed Circuit Heat Exchanger with Airfoil Fins Using High-Pressure Water", *International Conference on Micro/Nanoscale Heat Transfer. American Society of Mechanical Engineers, (ASME MNHMT)*, 2024, 88155: V001T09A001, *Oral presentation*.
2. **Liu W.**, Gu X., Liu Y., Wang J., Wang R., Fu Y.*, "Innovative Thermal Management for Liquid Hydrogen Aero Engines: A Comparison with Kerosene Systems", *11th European Conference for Aerospace Sciences, (EUCASS)*, 2025, *Oral presentation*.
3. Qi H., **Liu W.**, Shi S., Gang X., Fu Y.*, "Analysis of A Compact Printed Circuit Heat Exchanger with Airfoil Fins in Aero Engine Cooling Systems: An Experimental Study", *International Gas Turbine Congress, (IGTC)*, 2023.
4. Wang R., Liu Y., **Liu W.**, Fu Y.*, "Flow Method Measurement of Fluid Thermal Conductivity Based on Hot Wire", *3rd Asian Conference on Thermal Sciences, (ACTS)*, 2024.

REFERENCES

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Research collaborator

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Ph.D. co-advisor

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